

Even Sum

PLATFORM

DIFFICULTY

DATE

Platform: CodeChef

DIV 3

Solved: June 3, 2026

Problem

Remove exactly one element such that the sum of the remaining elements is even.

Algorithm

- Calculate total sum.
- If sum is even:
 - Need at least one even element.
- If sum is odd:
 - Need at least one odd element.

Key Observation

Let total sum be:

$$S = A_1 + A_2 + \dots + A_N$$

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After removing A_i :

$$\text{remaining sum} = S - A_i$$

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For this to be even:

$$S - A_i \equiv 0 \pmod{2}$$

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So S and A_i must have the same parity.

TC & SC

Time Complexity : $O(N)$

Space Complexity : $O(1)$

StatementSubmissionsClarifications

Even Sum

You are given an array A of N elements.

You must perform **exactly** one move. In this move, you have to choose an index i and remove A_i from the array.

Determine whether there exists an index i such that after removing A_i , the sum of the remaining elements is even.

Input Format

- The first line of input will contain a single integer T , denoting the number of test cases.
- Each test case consists of two lines of input.
 - The first line of each test case contains a single integer N , denoting the size of the array.
 - The second line of each test case contains N space-separated integers $A_1, A_2, \dots, A_N$, denoting the elements of the array.

Output Format

For each test case, output **Yes** if there is an index i such that after removing A_i , the sum of the remaining elements is even, otherwise output **No**.

You may print each character of the string in uppercase or lowercase (for example, the strings **yEs**, **yes**, **Yes**, and **YES** will all be treated as identical).

Constraints

- $1 \leq T \leq 100$
- $2 \leq N \leq 100$
- $1 \leq A_i \leq 100$

Sample 1:

Input	Output
4	No
2	Yes

Java

```
1 import java.util.*;
2 import java.lang.*;
3 import java.io.*;
4
5 class Codechef
6 {
7     public static void main (String[] args) throws java.lang.Exception
8     {
9         // your code goes here
10        Scanner sc = new Scanner(System.in);
11
12        int T = sc.nextInt();
13
14        while (T-- > 0) {
15            int N = sc.nextInt();
16
17            int sum = 0;
18            boolean hasEven = false, hasOdd = false;
19
20            for (int i = 0; i < N; i++) {
21                int x = sc.nextInt();
22                sum += x;
23
24                if (x % 2 == 0) hasEven = true;
25                else hasOdd = true;
26            }
27
28            if (sum % 2 == 0) {
29                System.out.println(hasEven ? "Yes" : "No");
30            } else {
31                System.out.println(hasOdd ? "Yes" : "No");
32            }
33        }
34    }
35
36    sc.close();
37 }
38
39
```

Test against Custom Input

4
2
5 7
3

Correct Answer

Submission ID: 1285408052

Visualize Code

Run

Submit